Notes: Eaton, Kortum, Neiman, and Romalis (AER, 2016)
Trade and the Global Recession

Contents

1 Big picture

- **Question.** Why did world manufacturing trade collapse much more sharply than GDP during the Great Recession, and which shocks explain the cross-country pattern of trade, manufacturing production, and GDP between 2008 and 2011?
- **Motivating fact.** Manufacturing trade plunged by roughly 29 percent from 2008:III to 2009:II, and trade relative to GDP fell by about 20 percent. Manufacturing production also fell, but less than trade.
- **Framework.** The paper embeds a multisector Eaton-Kortum style trade model into a dynamic multicountry real business cycle model with capital accumulation, input-output linkages, durable and nondurable manufactures, construction, services, trade frictions, and services deficits.
- **Shock accounting.** The model exactly accounts for observed paths of GDP, production, trade, and prices using shocks to bilateral trade frictions, sectoral productivity, investment efficiency, aggregate demand, nondurable demand, labor supply, and services deficits.
- **Main result.** A large decline in investment efficiency for durable manufacturing capital is the main driver of the global trade collapse. This shock shifts final spending away from highly traded durable manufacturing toward less traded sectors.
- **What matters less.** Shocks to trade frictions, productivity, and aggregate demand play secondary roles for the trade collapse, even though productivity and labor shocks matter more for real GDP.
- **Cross-country transmission.** The model identifies where shocks originate and how they propagate through the world trading system. US shocks mostly explain US outcomes, whereas Germany is much more exposed to shocks from the rest of the world.
- **Economic takeaway.** The Great Trade Collapse was less a story of rising trade barriers and more a story of sectoral demand composition, especially the collapse of demand for tradable durable investment goods.

2 Incremental contribution of the paper

- **From static trade accounting to dynamic global shock accounting.** Earlier quantitative trade models could map trade frictions and productivity into cross-sectional trade flows. This paper builds a dynamic framework that tracks capital accumulation and transition paths, allowing shocks today to affect future production and trade.
- **Multicountry, multisector, and dynamic in one system.** The paper combines a many-country gravity/trade framework with four broad sectors and two capital stocks. This is richer than two-country international macro models and richer than static multicountry gravity models.
- **Business-cycle accounting for the world economy.** The paper adapts the spirit of business-cycle accounting to a global setting. It recovers the realized shocks that exactly rationalize observed outcomes and then uses counterfactuals to measure which shocks matter.
- **Quantitative answer to the Great Trade Collapse debate.** The paper distinguishes trade-friction explanations, vertical-supply-chain explanations, sectoral-composition explanations, and productivity/demand explanations. It finds that durable investment-efficiency shocks dominate.
- **Sectoral demand composition as the central channel.** The incremental insight is that trade collapses disproportionately when demand shifts away from sectors that are both durable and highly traded. This does not require a large increase in measured trade barriers.
- **Country-origin decomposition.** The paper shows how shocks from one country propagate to other countries. This allows country-specific counterfactuals, such as US-only shocks and non-Germany shocks.

- **Distinction between trade, production, and GDP drivers.** The paper shows that the shocks explaining the trade collapse are not the same as the shocks explaining real GDP movements. Investment-efficiency shocks dominate trade and manufacturing production, while productivity and labor shocks matter more for real GDP.
- **Methodological contribution.** The model provides a reusable anatomy tool for studying global macroeconomic episodes with trade, capital accumulation, sectoral input-output relationships, and country-level shock propagation.

3 Data and measurement

3.1 Countries, sectors, and timing

- The model is calibrated to 21 economies: 20 named countries plus the Rest of World.
- Economic activity is divided into four sectors: construction, durable manufactures, nondurable manufactures, and services.
- Only durable and nondurable manufacturing are traded sectors in the model.
- Construction produces structures capital; durable manufacturing produces durable capital.
- The global recession window is 2008:III to 2009:II; the recovery window is 2009:II to 2011:I.

Interpretation: The sector split is crucial. The paper's main result depends on the fact that durable manufacturing is both investment-intensive and trade-intensive, so a shock to durable investment efficiency has a magnified effect on trade.

3.2 Observed variables

- The data include nominal and real GDP, sectoral production, prices, employment, investment, and bilateral trade by manufacturing sector.
- Trade in manufactures is measured as the average of exports and imports.
- Nominal variables are often expressed relative to global GDP to isolate relative movements and cross-country reallocations.
- Manufacturing trade is the focus because comprehensive bilateral trade data in services are unavailable and manufacturing accounts for most goods trade.

Interpretation: The data are structured to let the model match country-sector production and bilateral trade relationships, not just aggregate GDP.

3.3 Shock types

The model recovers six main country-specific shock types plus services-deficit shocks:

1. bilateral trade-friction shocks in durable and nondurable manufacturing;
2. sectoral productivity shocks in construction, durables, nondurables, and services;
3. investment-efficiency shocks for structures and durables capital;
4. aggregate demand shocks;
5. nondurable demand shocks;

6. labor supply shocks;
7. services deficit shocks, treated as transfers to respect accounting identities.

Interpretation: The decomposition is richer than a standard trade-cost exercise. It lets the authors ask whether the trade collapse is about trade barriers, sectoral productivity, demand composition, capital accumulation, or labor supply.

4 Models and theoretical specifications

4.1 Static production and trade block

The static block builds on Eaton and Kortum style trade. Each country-sector has a continuum of goods, sectoral CES aggregation, sector-specific productivity, intermediate-input use, and bilateral trade costs. Trade shares discipline bilateral trade frictions and technology.

- Countries buy each traded manufacturing good from the lowest-cost source.
- Sectoral price indexes depend on productivity, input costs, and trade frictions.
- Manufacturing sectors use inputs from all sectors, so shocks propagate through input-output links.
- A gravity structure links observed trade shares to bilateral trade costs and productivity.

4.2 Dynamic capital accumulation

There are two capital stocks, structures and durable equipment. Investment in sector k changes the capital stock with adjustment over time:

$$K^k_{n,t+1} = (1 - \delta^k)K^k_{n,t} + \lambda^k_{n,t+1} I^k_{n,t},$$

where $\lambda^k_{n,t+1}$ is investment efficiency.

Interpretation: Investment-efficiency shocks are the central mechanism. A fall in λ^D makes durable investment less effective, reducing final spending on durable manufacturing, which is intensively traded.

4.3 Households and demand

Households allocate expenditure across sectors and across time. Aggregate demand shocks and nondurable demand shocks shift the composition and level of final spending. Complete markets imply that global risk sharing pins down expenditure relationships across countries.

Interpretation: Demand shocks are backed out residually from spending patterns once technology, trade, and labor shocks are measured.

4.4 Shock extraction

The model is inverted so that a full set of shocks makes the dynamic solution exactly reproduce the observed data. In simplified terms:

- trade shares and prices identify trade-cost shocks;
- output and input prices identify productivity shocks;

- investment spending and capital dynamics identify investment-efficiency shocks;
- consumption composition identifies demand shocks;
- employment data identify labor shocks.

Interpretation: This is why the paper calls the exercise an anatomy. It does not estimate a few structural shocks statistically; it backs out shock paths that exactly reconcile the model with observed data.

4.5 Counterfactual accounting

The central counterfactual exercise solves the model with one subset of shocks active and all other shocks suppressed:

$$Y_t^{counterfactual}(S) = \text{model outcome with only shocks in set } S.$$

The contribution of shock group S is measured by how much the counterfactual reproduces the actual decline or recovery in trade, production, or GDP.

Interpretation: Counterfactuals are the identification device. The model tells us what would have happened if only trade frictions changed, or only durables investment efficiency changed, or only US shocks occurred.

5 Identification and empirical design

5.1 What identifies the main mechanism

- Durable goods are highly traded relative to nontradables and construction.
- The recession brought a large fall in durable investment efficiency.
- The model maps this fall into reduced final demand for durable manufactures.
- Because durable manufactures are highly traded, the resulting expenditure shift causes trade to fall more than GDP.

Interpretation: The main result is identified by the interaction of sectoral trade intensity and sectoral investment-efficiency shocks.

5.2 Why trade-friction explanations are limited

- Table 2 shows only modest average movements in trade frictions during the recession.
- Counterfactuals with trade-friction shocks alone do not reproduce most of the global trade collapse.
- Trade frictions matter more during the recovery than the collapse.

Interpretation: The paper does not deny that trade finance or protectionism changed. It argues that these changes are not quantitatively central for the global trade collapse in manufactures.

5.3 Country propagation design

- The authors run country-specific counterfactuals such as only US shocks or all shocks except Germany's.
- These counterfactuals isolate where shocks originate and how trade relationships transmit them.
- Countries with high trade exposure, such as Germany, are more affected by foreign shocks.

Interpretation: The framework distinguishes home-grown recessions from imported trade-linked recessions.

5.4 Limitations

- The model assumes perfect foresight and complete global asset markets.
- It abstracts from services trade, unemployment, financial frictions, imperfect competition, and uncertainty.
- Shocks are backed out from a finite episode rather than estimated as long-run stochastic processes.
- The counterfactuals are model-dependent, so their interpretation relies on the model's structure.

Interpretation: These limitations are important. The paper is a transparent accounting framework, not a final model of all crisis mechanisms.

6 Section-by-section study guide

- **Introduction.** Motivates the 29 percent manufacturing trade collapse and previews the dynamic multicountry model.
- **Related literature.** Positions the paper relative to trade-collapse explanations, vertical supply chains, trade credit, investment-specific shocks, and multicountry macro-trade models.
- **A first look at the data.** Shows that trade and manufacturing production fell sharply after 2008:III, while construction declined later and recovered weakly.
- **Model.** Develops the dynamic multicountry multisector equilibrium with trade, input-output linkages, capital accumulation, and complete markets.
- **Recovering shocks.** Explains how trade frictions, productivity, investment efficiency, demand, labor, and services deficits are backed out of the data.
- **Counterfactuals.** Uses subsets of shocks to explain the global and country-level decline/recovery of trade, production, and GDP.
- **Conclusion.** Interprets the trade collapse as mainly a demand-composition shift caused by falling durable investment efficiency.

7 Figures and tables

7.1 Figure 1 and Figure 2: Global dynamics and cross-country first look

Read:

- Figure 1 shows global trade, manufacturing production, and construction relative to GDP from 2000 onward.
- Trade and production fall sharply after 2008:III and rebound after 2009:II; construction declines later and recovers weakly.
- Figure 2 shows country-sector changes during the recession and recovery.
- Durables trade and production fall more sharply than nondurables in the recession.

Detailed interpretation: Figure 1 establishes the puzzle: trade collapses more than GDP. Figure 2 shows the cross-country and sectoral pattern behind the aggregate fact. Countries with large durable manufacturing contractions also have large trade contractions, which motivates the model's focus on durable investment efficiency.

Economic message: The Great Trade Collapse is not simply an aggregate demand collapse. It is a sectoral-composition event concentrated in durable manufacturing, a highly traded sector.

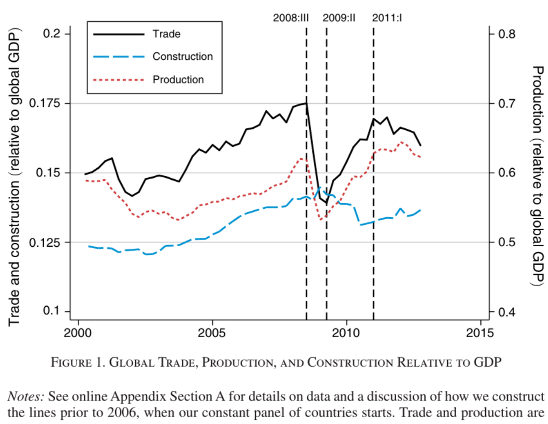

Figure 1: Global trade, production, and construction relative to GDP

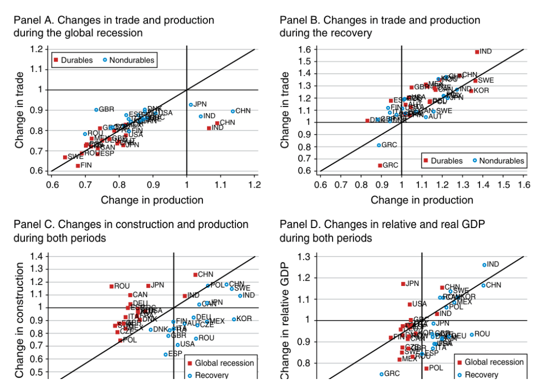

Figure 2: Trade, production, construction, and GDP

7.2 Table 1 and Figure 3: Country base-period exposure and capital-stock dynamics

Read:

- Table 1 summarizes country shares of GDP, trade, trade/GDP, production/GDP, and construction/GDP in 2008:III.
- Smaller open economies have much higher trade/GDP ratios, so their production declines have large effects on global trade.
- Figure 3 shows the capital-stock transition paths for structures and durables in the United States, China, and the Rest of World.
- Durables capital adjusts faster and is more volatile than structures capital.

Detailed interpretation: Table 1 explains the between-country composition channel: countries with higher trade intensities matter disproportionately for global trade. Figure 3 explains why dynamic capital accumulation matters: shocks today shift capital paths far beyond the observed recession window.

Economic message: Trade exposure and capital dynamics are both amplification mechanisms. Trade-weighted country composition amplifies production changes; durable capital dynamics amplify investment-efficiency shocks.

TABLE 1—SUMMARY STATISTICS FOR 20 COUNTRIES AND REST OF WORLD, 2008:III (*Percent*)

	Country	Code	Share of global GDP	Share of global trade	Trade/ GDP	Production/ GDP	Construction/ GDP
1.	Austria	AUT	0.7	1.5	39.5	57.8	16.0
2.	Canada	CAN	2.6	3.0	19.9	40.0	57.6
3.	China	CHN	7.5	10.3	24.0	153.1	18.1
4.	Czech Republic	CZE	0.4	1.3	56.7	95.9	63.5
5.	Denmark	DNK	0.6	0.9	28.0	37.6	83.1
6.	Finland	FIN	0.5	0.8	29.2	71.2	44.7
7.	France	FRA	4.7	5.3	19.7	42.8	54.0
8.	Germany	DEU	6.0	11.1	32.4	68.2	48.1
9.	Greece	GRC	0.6	0.4	13.4	31.5	55.4
10.	India	IND	2.1	1.6	13.3	56.1	28.8
11.	Italy	ITA	3.8	4.4	20.1	69.6	31.6
12.	Japan	JPN	7.4	5.3	12.5	71.7	19.6
13.	Mexico	MEX	1.9	2.4	21.4	55.2	55.9
14.	Poland	POL	0.9	1.6	29.6	65.6	53.8
15.	Romania	ROU	0.3	0.5	27.4	48.9	66.4
16.	South Korea	KOR	1.6	3.2	36.0	138.2	28.4
17.	Spain	ESP	2.6	2.7	17.7	50.1	39.8
18.	Sweden	SWE	0.8	1.4	30.5	57.7	53.5
19.	United Kingdom	GBR	4.5	4.2	16.6	34.4	58.2
20.	United States	USA	23.9	12.3	9.0	37.1	32.0
21.	Rest of World	ROW	26.5	25.7	17.0	61.6	30.4

Notes: Trade and production data are for manufactures. Trade is defined as the average of exports and imports. Trade data do not include flows between countries within Rest of World. See online Appendix Section A for details.

Table 1: Summary statistics for countries

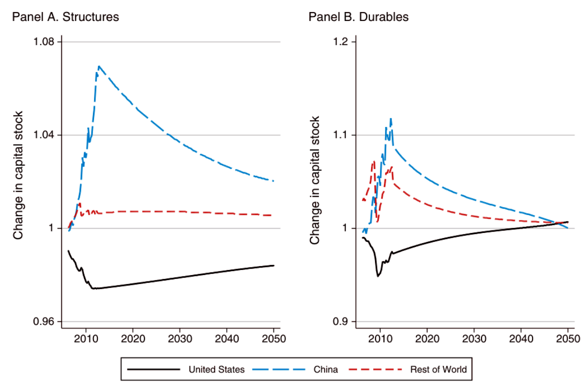

Figure 3: Capital stocks during transition

7.3 Table 2 and Table 3: Trade frictions, services deficits, labor, and productivity

Read:

- Table 2 reports trade-friction shocks, services deficits, and labor shocks over the pre-recession, recession, and recovery periods.
- Average trade-friction movements are modest during the recession; labor falls for many countries.
- Table 3 reports productivity shocks by sector.
- Construction and services productivity fall during the recession, while manufacturing productivity patterns are not the main source of the trade collapse.

Detailed interpretation: These tables rule in and rule out candidate explanations. Trade barriers do not rise enough on average to explain the collapse. Productivity matters for GDP, especially in nontraded sectors, but is not the central driver of the relative decline in trade.

Economic message: The crisis was not primarily a global protectionism episode, nor primarily a traded-sector productivity collapse. The decisive shock lies elsewhere.

TABLE 2—TRADE FRICTIONS, SERVICES DEFICITS, AND LABOR SUPPLY SHOCKS

	$\hat{d}_n^D$			$\hat{d}_n^N$			$\Delta \frac{D_n^S}{Y_n}$			$\hat{L}_n$		
	Prior period	Global recession	Recov period	Prior period	Global recession	Recov period	Prior period	Global recession	Recov period	Prior period	Global recession	Recov period
World	**0.991**	**0.992**	**0.948**	**0.980**	**0.973**	**0.972**	**0.000**	**0.000**	**0.000**	**1.010**	**0.983**	**0.999**
Austria	0.999	1.114	0.933	0.976	0.959	1.029	0.000	−0.018	0.006	1.012	0.994	1.007
Canada	1.005	1.060	0.944	0.987	1.042	0.988	−0.002	0.048	−0.002	1.019	0.982	1.010
China	0.990	1.028	0.935	0.999	1.002	0.933	0.014	−0.017	0.032	1.004	1.001	1.002
Czech R.	0.942	0.959	0.908	0.968	0.950	0.941	0.007	−0.033	0.011	1.007	0.994	0.991
Denmark	0.981	1.052	0.732	0.971	0.892	0.951	0.001	0.032	−0.004	1.008	0.981	0.976
Finland	0.996	1.124	1.004	0.991	1.033	0.951	0.000	−0.005	0.001	1.011	0.975	0.993
France	1.019	0.970	0.984	0.992	0.968	0.983	0.003	−0.014	0.008	1.008	0.992	0.999
Germany	0.984	0.993	0.977	0.974	0.977	0.985	0.000	−0.025	0.003	1.007	1.010	1.004
Greece	1.024	0.963	1.033	0.998	1.017	1.011	0.008	−0.020	0.003	1.015	0.987	0.973
India	0.959	1.100	0.936	0.958	1.044	0.966	−0.001	−0.014	0.015	1.018	1.001	1.003
Italy	0.994	1.004	0.982	0.985	0.972	0.968	0.003	−0.017	0.004	1.014	0.984	0.994
Japan	1.016	0.927	0.951	0.970	0.996	0.996	0.003	−0.041	0.012	0.999	0.988	0.994
Mexico	0.979	0.941	0.881	0.991	0.964	0.987	0.001	0.006	−0.002	1.020	1.009	1.015
Poland	0.953	0.903	0.974	0.964	0.886	0.980	0.004	−0.017	−0.003	1.009	1.018	0.982
Romania	0.952	0.911	0.894	0.975	0.915	0.937	0.005	−0.015	−0.015	1.007	0.974	0.991
S. Korea	0.994	0.916	0.977	0.996	0.917	0.949	0.011	−0.101	0.040	1.014	0.995	1.011
Spain	0.995	1.039	0.926	0.977	0.954	0.971	0.003	−0.028	0.006	1.037	0.920	0.977
Sweden	0.995	1.006	0.950	0.979	0.982	0.998	−0.004	−0.008	0.008	1.012	0.984	1.000
UK	0.999	0.990	0.897	0.984	0.868	0.958	0.000	−0.003	−0.002	1.009	0.985	0.998
US	0.991	1.004	0.939	0.983	0.985	0.982	0.002	−0.026	0.002	1.010	0.960	0.990
RoW	0.989	0.991	0.963	0.979	0.983	0.973	−0.007	0.062	−0.023	1.012	0.994	1.007

Notes: Global recession is 2008:III to 2009:II. Prior Period begins in 2000:I with exceptions documented in online Appendix Section A. Recovery period is 2009:II to 2011:I. Shocks are annualized. Trade friction shocks are calculated as a trade-weighted average of the bilateral shocks (see footnote 28). The services trade deficit shocks are the changes in the deficits divided by GDP at the beginning of the quarter, averaged over the period.

Table 2: Trade frictions, services deficits, and labor supply shocks

TABLE 3—PRODUCTIVITY SHOCKS

	$\hat{A}_n^C$			$\hat{A}_n^D$			$\hat{A}_n^N$			$\hat{A}_n^S$		
	Prior period	Global recession	Recov period	Prior period	Global recession	Recov period	Prior period	Global recession	Recov period	Prior period	Global recession	Recov period
World	**0.995**	**0.934**	**1.043**	**1.011**	**1.030**	**1.009**	**1.006**	**1.057**	**1.007**	**1.012**	**0.962**	**1.019**
Austria	1.008	0.940	0.988	1.023	1.141	0.916	0.995	0.998	1.041	1.010	0.951	1.026
Canada	0.989	0.778	1.078	1.031	0.979	1.002	1.014	0.958	1.027	1.009	1.000	1.013
China	1.033	0.845	1.059	1.064	1.124	1.013	1.074	1.131	1.014	1.020	0.955	1.026
Czech R.	0.995	1.125	0.917	0.992	0.937	0.918	1.006	0.984	0.943	1.014	0.943	1.035
Denmark	0.996	1.003	1.063	0.982	1.061	0.631	0.983	0.892	0.940	1.013	0.963	1.025
Finland	1.004	0.879	1.121	1.011	1.018	1.002	0.998	1.018	0.976	1.010	0.915	1.019
France	0.967	0.995	1.018	1.010	0.983	0.989	1.006	1.056	0.989	1.017	0.967	1.025
Germany	0.997	0.979	1.028	0.995	0.979	1.004	0.985	1.013	0.991	1.008	0.925	1.036
Greece	1.021	0.983	1.004	1.016	0.999	1.024	1.009	1.031	0.997	1.030	0.969	0.974
India	1.007	0.909	1.059	1.013	1.111	1.042	1.026	1.013	1.044	1.063	1.060	1.056
Italy	0.995	0.920	0.986	1.000	1.021	0.981	1.000	1.016	0.978	1.002	0.942	1.028
Japan	1.006	0.917	0.952	0.990	1.031	0.993	0.978	1.040	0.989	1.016	0.924	1.032
Mexico	1.012	0.970	1.037	1.008	0.932	0.926	1.016	0.945	1.015	1.025	0.969	1.056
Poland	0.969	1.359	0.963	1.013	1.011	1.019	1.019	1.004	1.012	1.024	0.983	1.035
Romania	0.951	0.864	1.008	0.971	0.907	0.899	0.981	0.872	0.940	1.114	1.024	1.072
S. Korea	0.989	0.979	1.027	1.000	1.043	0.996	1.040	1.051	0.994	1.008	0.947	1.020
Spain	0.984	0.967	1.010	1.005	1.057	0.954	1.003	0.996	0.978	1.012	0.998	1.021
Sweden	0.988	0.885	1.021	1.004	0.948	1.022	0.991	1.008	1.038	1.016	0.965	1.027
UK	0.990	0.901	1.071	1.007	0.980	0.927	0.998	0.873	0.978	1.008	0.991	1.007
US	0.979	0.995	1.029	1.003	1.039	0.988	0.978	1.205	0.940	1.018	0.955	1.036
RoW	1.002	0.918	1.067	1.024	0.981	1.050	1.025	0.982	1.059	1.003	0.963	0.987

Notes: Global recession is 2008:III to 2009:II. Prior period begins in 2000:I with exceptions documented in online Appendix Section A. Recovery period is 2009:II to 2011:I. Shocks are annualized. Productivity shocks for World are aggregated across countries, analogous to trade frictions in Table 2.

Table 3: Productivity shocks

7.4 Table 4 and Table 5: Investment-efficiency/demand shocks and recession accounting

Read:

- Table 4 shows the dramatic decline in durables investment efficiency during the recession.
- The world durables investment-efficiency shock falls sharply, while construction investment-efficiency changes are milder.
- Table 5 reports counterfactual trade outcomes with different shock groups active.
- Durables investment-efficiency shocks reproduce most of the global trade collapse; trade friction, productivity, labor, and services-deficit shocks contribute much less.

Detailed interpretation: Table 4 identifies the dominant shock, and Table 5 shows its quantitative contribution. The key column is the counterfactual with durables investment-efficiency shocks, which generates a large decline in trade relative to GDP.

Economic message: The trade collapse is mainly a collapse in demand for tradable durable investment goods caused by a fall in the efficiency of investing in durable capital.

TABLE 4—INVESTMENT EFFICIENCY AND DEMAND SHOCKS

	$\hat{\chi}_n^C$			$\hat{\chi}_n^D$			$\hat{\psi}_n^N$			$\hat{\phi}_n$		
	Prior period	Global recession	Recov period	Prior period	Global recession	Recov period	Prior period	Global recession	Recov period	Prior period	Global recession	Recov period
World	**1.007**	**0.984**	**0.976**	**0.993**	**0.772**	**1.076**	**0.995**	**0.772**	**1.104**	**1.000**	**1.000**	**1.000**
Austria	1.021	0.958	0.963	1.015	0.869	1.005	1.003	0.803	1.236	1.003	0.969	0.963
Canada	1.040	1.103	1.012	0.949	0.693	1.067	0.957	0.974	0.917	1.027	0.894	1.081
China	1.077	1.244	1.048	1.043	1.042	1.098	1.012	1.041	1.078	1.132	1.162	1.094
Czech R.	1.088	0.697	1.017	1.023	0.563	1.115	0.936	0.779	1.091	1.093	0.876	0.955
Denmark	1.018	0.863	0.909	1.000	0.646	0.816	0.982	0.636	0.939	1.017	0.902	0.978
Finland	1.032	0.902	0.908	1.032	0.666	1.070	1.018	0.730	0.961	1.020	0.975	0.949
France	1.062	0.900	0.926	1.002	0.724	1.017	0.990	0.750	1.071	1.011	0.945	0.960
Germany	0.984	0.966	0.954	1.013	0.741	1.040	0.998	0.809	1.121	0.995	0.958	0.945
Greece	1.003	0.962	0.729	1.030	0.742	0.902	0.984	0.793	1.084	1.054	0.944	0.885
India	1.033	1.035	1.045	1.029	0.848	1.105	1.022	1.045	0.954	1.028	0.933	1.177
Italy	1.031	0.939	0.945	1.007	0.634	1.060	0.985	0.675	1.099	1.013	1.009	0.928
Japan	0.926	1.228	1.052	0.959	0.884	1.028	1.020	0.726	1.212	0.926	1.255	1.004
Mexico	1.005	0.747	0.977	0.963	0.581	1.026	1.001	0.954	1.081	1.000	0.742	1.080
Poland	1.097	0.533	1.120	1.013	0.519	1.052	0.992	0.945	1.164	1.063	0.675	1.034
Romania	1.202	1.044	0.918	1.088	0.454	1.078	0.965	0.662	1.109	1.138	0.684	0.959
S. Korea	1.009	0.870	0.993	1.038	0.628	1.231	1.035	0.750	1.052	0.994	0.852	1.069
Spain	1.096	0.983	0.854	0.968	0.414	1.095	0.959	0.690	1.137	1.048	0.915	0.951
Sweden	1.020	0.814	1.075	0.994	0.533	1.185	1.014	0.955	1.049	1.003	0.821	1.083
UK	1.020	0.854	0.874	0.982	0.747	0.997	0.983	0.736	0.927	0.998	0.787	0.980
US	0.974	0.970	0.868	0.953	0.787	1.029	0.997	0.709	1.109	0.982	1.061	0.956
RoW	1.031	0.911	0.991	1.030	0.723	1.107	0.990	0.721	1.127	1.036	0.977	1.042

Notes: Global recession is 2008:III to 2009:II. Prior period begins in 2000:I with exceptions documented in online Appendix Section A. Recovery period is 2009:II to 2011:I. Shocks are annualized. Shocks to investment efficiency for the World are calculated as an investment-weighted average of the country shocks (see footnote 31). Shocks to

Table 4: Investment efficiency and demand shocks

TABLE 5—TRADE DURING THE GLOBAL RECESSION

	Trade/ world GDP in 2008:III (percent)	All shocks (i.e., data)	Change $\left(\frac{2009:II}{2008:III}\right)$ in trade in various counterfactuals: Trade friction shocks $\left(\hat{d}_{ni}^j\right)$	Prod. shocks $\left(\hat{A}_n^j\right)$	Inv. efficy. in structures shocks $\left(\hat{\chi}_n^C\right)$	Inv. efficy. in durables shocks $\left(\hat{\chi}_n^D\right)$	Nondurables demand shocks $\left(\hat{\psi}_n^N\right)$	Aggregate demand shocks $\left(\hat{\phi}_n\right)$	Services deficits shocks $\left(D_n^S\right)$	Labor supply shocks $\left(\hat{L}_n\right)$
World	**17.5**	**0.795**	**0.977**	**1.001**	**0.994**	**0.868**	**0.964**	**0.999**	**0.996**	**0.996**
Austria	0.3	0.790	0.901	1.040	0.990	0.870	0.963	0.992	0.991	0.993
Canada	0.5	0.752	0.910	0.991	1.016	0.868	0.988	1.029	1.011	1.006
China	1.8	0.852	0.915	1.103	0.981	0.926	0.971	0.976	0.992	0.994
Czech R.	0.2	0.746	1.020	0.908	0.996	0.809	0.968	1.003	0.989	0.990
Denmark	0.2	0.805	1.010	0.960	0.992	0.857	0.948	1.010	1.004	0.991
Finland	0.1	0.675	0.863	0.974	0.999	0.846	0.972	1.007	0.999	0.996
France	0.9	0.828	1.033	0.979	0.996	0.864	0.955	1.000	0.995	0.996
Germany	1.9	0.784	0.990	0.960	0.992	0.858	0.965	0.999	0.991	0.998
Greece	0.1	0.834	1.020	1.000	1.000	0.895	0.956	0.977	1.002	1.000
India	0.3	0.831	0.913	1.063	1.002	0.916	0.981	1.012	0.992	1.004
Italy	0.8	0.770	0.986	0.983	0.995	0.850	0.948	0.991	0.996	0.994
Japan	0.9	0.773	0.971	0.997	0.971	0.886	0.953	0.954	0.975	0.981
Mexico	0.4	0.774	1.014	0.975	0.998	0.811	0.987	1.033	1.000	1.002
Poland	0.3	0.758	0.998	0.993	0.982	0.804	0.975	1.027	0.994	0.998
Romania	0.1	0.719	1.072	0.955	0.999	0.754	0.953	1.000	0.995	0.991
S. Korera	0.6	0.832	0.997	1.027	0.994	0.867	0.971	1.007	0.980	0.994
Spain	0.5	0.756	0.975	1.010	0.995	0.827	0.952	0.993	0.997	0.986
Sweden	0.3	0.724	0.946	0.943	0.993	0.824	0.974	1.023	0.994	0.991
UK	0.7	0.809	1.029	0.978	0.995	0.861	0.959	1.015	0.996	0.995
US	2.2	0.808	0.984	1.013	0.998	0.869	0.963	1.002	0.999	0.995
RoW	4.5	0.788	0.977	0.992	0.999	0.872	0.962	1.002	1.002	1.000

Table 5: Trade during the global recession

7.5 Figure 4 and Figure 5: Global trade counterfactuals and cross-sectional trade variation

Read:

- Figure 4 plots actual global trade and counterfactual paths with different shocks active.
- The durables investment-efficiency counterfactual tracks much of the trade decline.
- Figure 5 plots country-level trade declines against shock-specific counterfactual declines.
- Durables investment-efficiency shocks have the strongest cross-country explanatory power for trade.

Detailed interpretation: Figure 4 is the time-series accounting figure; Figure 5 is the cross-country accounting figure. Together they show that the same shock explains both the global aggregate movement and much of the country-level variation.

Economic message: A convincing explanation of the Great Trade Collapse must explain both the aggregate time path and cross-country heterogeneity. Durables investment efficiency does both.

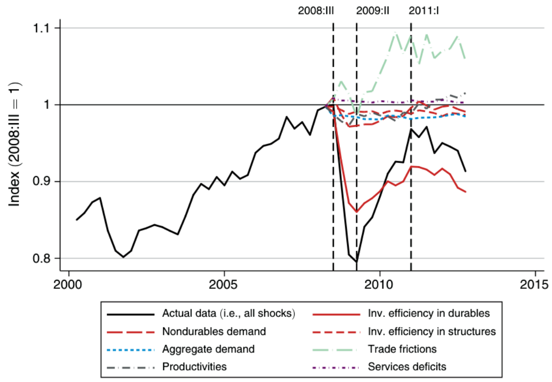

FIGURE 4. ACTUAL AND COUNTERFACTUAL EVOLUTION OF GLOBAL TRADE

Lines beginning in 2008:III represent counterfactual outcomes with the paths of indicated shocks at their

Figure 4: Actual and counterfactual evolution of global trade

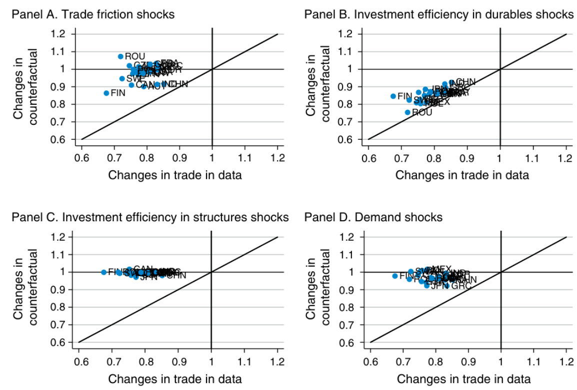

FIGURE 5. CROSS-SECTIONAL EXPLANATORY POWER OF VARIOUS SHOCKS FOR TRADE DURING THE GLOBAL RECESSION

Figure 5: Cross-sectional explanatory power for trade

7.6 Figure 6 and Figure 7: Production and GDP counterfactuals

Read:

- Figure 6 repeats the shock-accounting exercise for manufacturing production.
- Investment-efficiency shocks in durables again explain a large share of production declines.
- Figure 7 repeats the exercise for nominal relative GDP.
- Demand shocks explain more of relative GDP than they explain trade or manufacturing production.

Detailed interpretation: These figures show that different outcomes have different shock drivers. Trade and production are closely tied to durable investment efficiency, while relative GDP responds more to aggregate demand and other macro forces.

Economic message: The paper's accounting is multidimensional: the right explanation depends on whether the target is trade, production, nominal relative GDP, or real GDP.

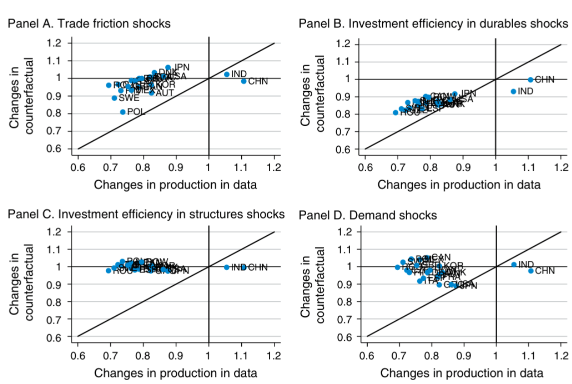

FIGURE 6. CROSS-SECTIONAL EXPLANATORY POWER OF VARIOUS SHOCKS FOR PRODUCTION DURING THE GLOBAL RECESSION

Figure 6: Cross-sectional explanatory power for production

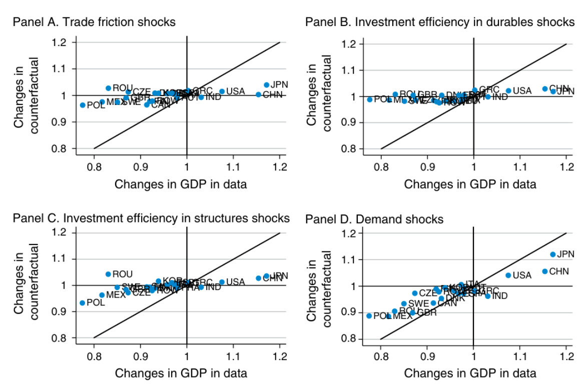

FIGURE 7. CROSS-SECTIONAL EXPLANATORY POWER OF VARIOUS SHOCKS FOR GDP DURING THE GLOBAL RECESSION

Figure 7: Cross-sectional explanatory power for GDP

7.7 Figure 8 and Figure 9: Trade/GDP and real GDP mechanisms

Read:

- Figure 8 shows that investment-efficiency and demand shocks explain most of the cross-country variation in trade/GDP declines.
- Figure 9 shows that productivity and labor shocks explain cross-country variation in real GDP declines.
- The distinction between trade/GDP and real GDP is important because real activity is driven by different forces than trade intensity.

Detailed interpretation: Figure 8 summarizes the mechanism for the relative trade collapse, while Figure 9 explains why real GDP requires a different shock bundle. A model of trade alone would miss this distinction.

Economic message: The trade collapse was not simply the real GDP recession magnified. Trade/GDP and real GDP are different objects with different shock decompositions.

The figures plot, against the actual changes in a country's GDP during the three quarter recessi
es that occur over that period in a counterfactual exposed only to the path of the indicated shocks. All
ative to global GDP.

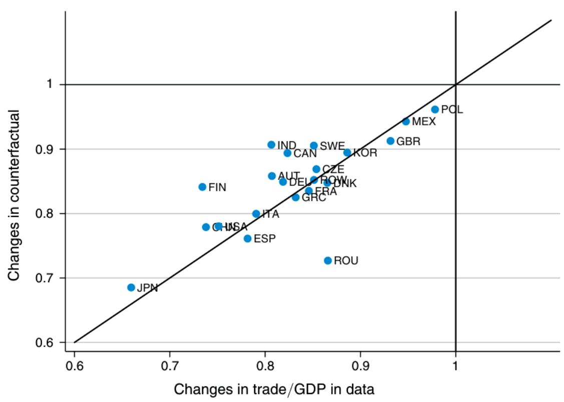

FIGURE 8. EXPLANATORY POWER OF INVESTMENT EFFICIENCY AND DEMAND SHOCKS FOR TRADE/GDP DURING THE GLOBAL RECESSION

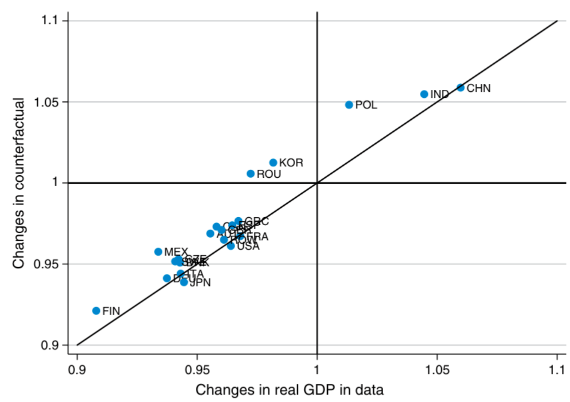

FIGURE 9. EXPLANATORY POWER OF LABOR AND PRODUCTIVITY SHOCKS FOR REAL GDP DURING THE GLOBAL RECESSION

The figure plots, against the actual changes in a country's real GDP during the three quarter recessi

Figure 8: Investment efficiency/demand shocks for trade/GDP

Figure 9: Labor and productivity shocks for real GDP

7.8 Figure 10 and Figure 11: Country-origin counterfactuals

Read:

- Figure 10 isolates the effect of US shocks on all countries.
- US shocks explain much of US outcomes, but they do not fully explain the global recession.
- Figure 11 removes Germany's own shocks and leaves shocks from the rest of the world active.
- Germany remains strongly affected, showing that foreign shocks matter for a highly trade-exposed economy.

Detailed interpretation: These counterfactuals identify where shocks originate and how they propagate. The United States is relatively domestically driven; Germany is much more exposed to foreign shocks. The results illustrate why a multicountry framework is necessary.

Economic message: Trade is a transmission mechanism. Global recessions are not just synchronized domestic shocks; they are also cross-border propagation through production and trade networks.

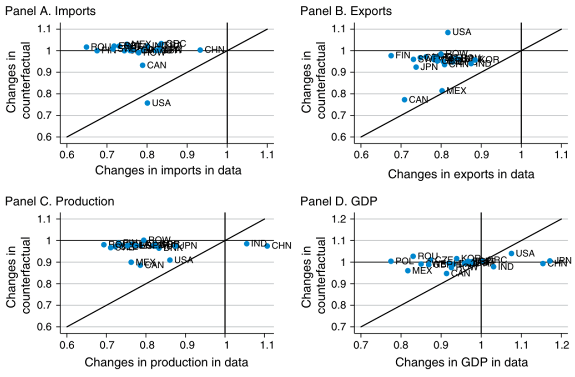

FIGURE 10. COUNTERFACTUAL RESPONSE TO US SHOCKS DURING THE GLOBAL RECESSION

Figure 10: Counterfactual response to US shocks

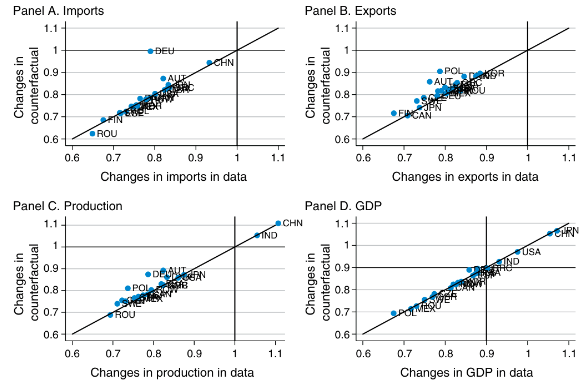

FIGURE 11. COUNTERFACTUAL RESPONSE TO SHOCKS OTHER THAN GERMANY'S DURING THE GLOBAL RECESSION

Figure 11: Counterfactual response to non-Germany shocks

8 Detailed result map

- **Fact 1:** Manufacturing trade falls far more than GDP from 2008:III to 2009:II.
- **Fact 2:** Durable manufacturing is more traded and falls more sharply than nondurables.
- **Mechanism 1:** A fall in durable investment efficiency shifts final spending away from durable manufacturing.
- **Mechanism 2:** Because durables are highly traded, spending shifts create disproportionate trade declines.
- **Accounting result:** Durables investment-efficiency shocks explain most of the trade collapse relative to GDP.
- **Secondary forces:** Trade frictions and productivity matter less for the trade collapse, but trade frictions matter in the recovery.
- **GDP result:** Real GDP movements are better explained by productivity and labor shocks.
- **Propagation result:** Shocks originate in countries and sectors and transmit differently depending on trade exposure.

9 Short summary

- The paper studies the collapse of world manufacturing trade during the Great Recession.
- It builds a dynamic multicountry, multisector general equilibrium model with trade, input-output linkages, and capital accumulation.
- It recovers shocks that exactly rationalize country-sector paths of trade, production, GDP, and prices.
- The main shock is a sharp decline in durables investment efficiency.
- This shock reduces final spending on durable manufacturing, a highly traded sector.
- Trade falls more than GDP because demand shifts away from traded goods.
- Trade frictions and productivity shocks are not the main causes of the trade collapse.

- Productivity and labor shocks are more important for real GDP than for trade.
- The model also measures how shocks from one country transmit to others.
- The paper's main contribution is a global dynamic trade-macro accounting framework for crisis episodes.

10 Conclusion

10.1 Core conclusion

Eaton, Kortum, Neiman, and Romalis show that the Great Trade Collapse was driven primarily by a sharp decline in the efficiency of investment in durable manufacturing capital. This shock shifted global final spending away from tradable durable goods and toward less traded sectors, causing trade to fall much more sharply than GDP.

10.2 Economic intuition

Durable manufactures are highly traded and closely tied to investment spending. When investment in durables becomes less efficient, households and firms reduce spending on durable capital goods. Since these goods are traded intensively, the decline shows up as a large trade collapse. The mechanism does not require a large increase in trade barriers.

10.3 Incremental contribution revisited

The paper's incremental contribution is to integrate dynamic capital accumulation into a multicountry, multisector trade model and to use that model as a shock-accounting device. It does not merely document the trade collapse; it decomposes it into shock types and origins.

10.4 Policy and research implications

The findings suggest that policies focused only on trade finance or protectionism miss much of the mechanism behind the Great Trade Collapse. More broadly, international macro models should track sectoral expenditure composition and capital accumulation when explaining trade dynamics.

10.5 Limitations

The model abstracts from financial frictions, uncertainty, imperfect competition, services trade, oil and agriculture trade, and unemployment. These simplifications make the accounting transparent but leave important extensions for future work.

10.6 Takeaway

The central takeaway is that trade collapses when demand shifts away from traded sectors. During the Great Recession, the critical shift was away from durable investment goods, caused by a sharp fall in durable investment efficiency.